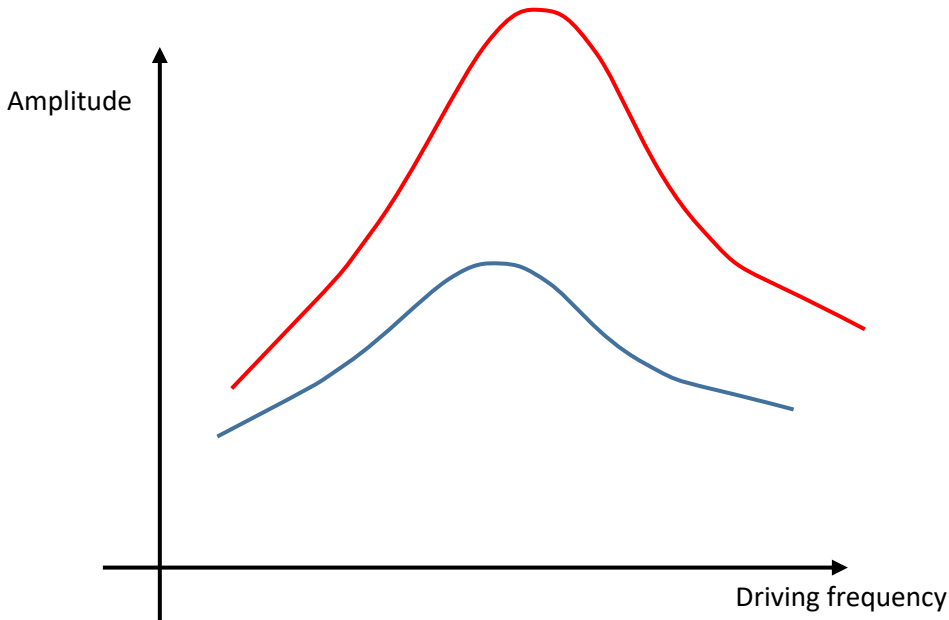


3		(i)	Resonance	[B1]
		(ii)	the bridge can swing on its own natural frequency. When the frequency of the forced oscillation due to walking pedestrians matches the natural frequency of the bridge, (resonance occurs). The amplitude may increase and damage the bridge structure.	[B1] [B1] [B1]
		(iii)	 Fig. 1.1	
			Correct shape and amplitude. Since it is before damping, the amplitude must be higher. Damping will make the period longer, thus frequency is lower. So before damping, the frequency must be higher.	[B1] [B1]